Polynomials and polynomial functions modulo p

Question 1

Give an example of a homogeneous polynomial of degree 4 in 3 variables over $\mathbb{Z}/\langle 2\rangle$ that has no *non-trivial* solutions.

Solution. The polynomial needs to be non-zero at all points

Let x, y and z be the variables and $f(x, y, z)$ be the polynomial.

In order that $f(1, 0, 0) \neq 0$ we need the coefficient of x^4 in f to be non-zero, which means 1!

Similarly, the coefficients of y^4 and z^4 have to be 1.

In order that $f(1, 1, 0) \neq 0$ we need at least one more term since $x^4 + y^4 + z^4$ is 0 at this point. For example, we can take the coefficient of x^2y^2 to be 1.

Similarly, we can take the coefficients of y^2z^2 and z^2x^2 to be 1.

Now, $x^4 + y^4 + z^4 + x^2y^2 + y^2z^2 + z^2x^2$ is a polynomial which is non-zero at all the points with at least one coordinate 0. However, it is 0 at $(1, 1, 1)$! So we need one term which is non-zero at this point but 0 at all other points; for example, we can take the term x^2yz.

Thus $f(x, y, z) = x^4 + y^4 + z^4 + x^2y^2 + y^2z^2 + z^2x^2 + x^2yz$ is a polynomial of the required type.

Question 2

Consider the polynomial $h(x, y) = 1 + x^4y + xy^4$ over $\mathbb{Z}/\langle 3\rangle$.

Find a polynomial $g(x, y)$ whose total degree is less than or equal to 4 which takes the same values of $h(x, y)$ at points of $(\mathbb{Z}/\langle 3\rangle)^2$.

Solution. We look for a representative $g(x, y)$ for the image of $h(x, y)$ $R = (\mathbb{Z}/\langle 3\rangle)[x, y]/(x^3 - x, y^3 - y)$. Every polynomial in R is uniquely determined by its associated function.

We note that

$$h(x, y) = 1 + x^4y + xy^4 = 1 + x^2y + xy^2 \pmod{\langle x^3 - x, y^3 - x\rangle}$$

Thus, $g(x, y) = 1 + x^2y + xy^2$ is the required polynomial.